

OpenClaw AI Script Peripheral Control Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the "having AI generate independently runnable scripts" part.

1. Tutorial Objectives

This article demonstrates that OpenClaw can not only directly control peripherals but also help you save control logic into script files through the approach of "natural language generation of hardware scripts". After completing this tutorial, readers will be able to:

- Understand the basic approach to generating hardware control scripts with OpenClaw
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Generate four types of common scripts: RGB, servo, temperature and humidity, OLED

2. Hardware Preparation

- OpenClaw main control board
- Servo / RGB light strip / OLED / DHT temperature and humidity sensor
- `[Optional] AI voice module + speaker`

Note:

- This tutorial does not require connecting all peripherals at once
- If you want to demonstrate which type of script, just connect the corresponding hardware first

3. Software Preparation

Code entry points used in this tutorial:

- Script generation skill: `/home/pi/.openclaw/workspace/skills/script-runner/SKILL.md`
- Script default save directory: `/home/pi/Desktop/`
- Hardware command example directory: `/home/pi/hardware_hal_demo/`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

The core of this capability is not "execute an immediate command" but having OpenClaw help you generate an independently runnable Python script. According to existing skill conventions:

- When users say "generate script", "write a script", "save to desktop", it will preferentially use `script-runner`
- Hardware scripts use `openclaw_lib` by default
- Default serial port is `/dev/ttyAMA0`
- Default baud rate is `115200`
- Scripts are usually saved to `/home/pi/Desktop/`

You can think of it as two usage methods:

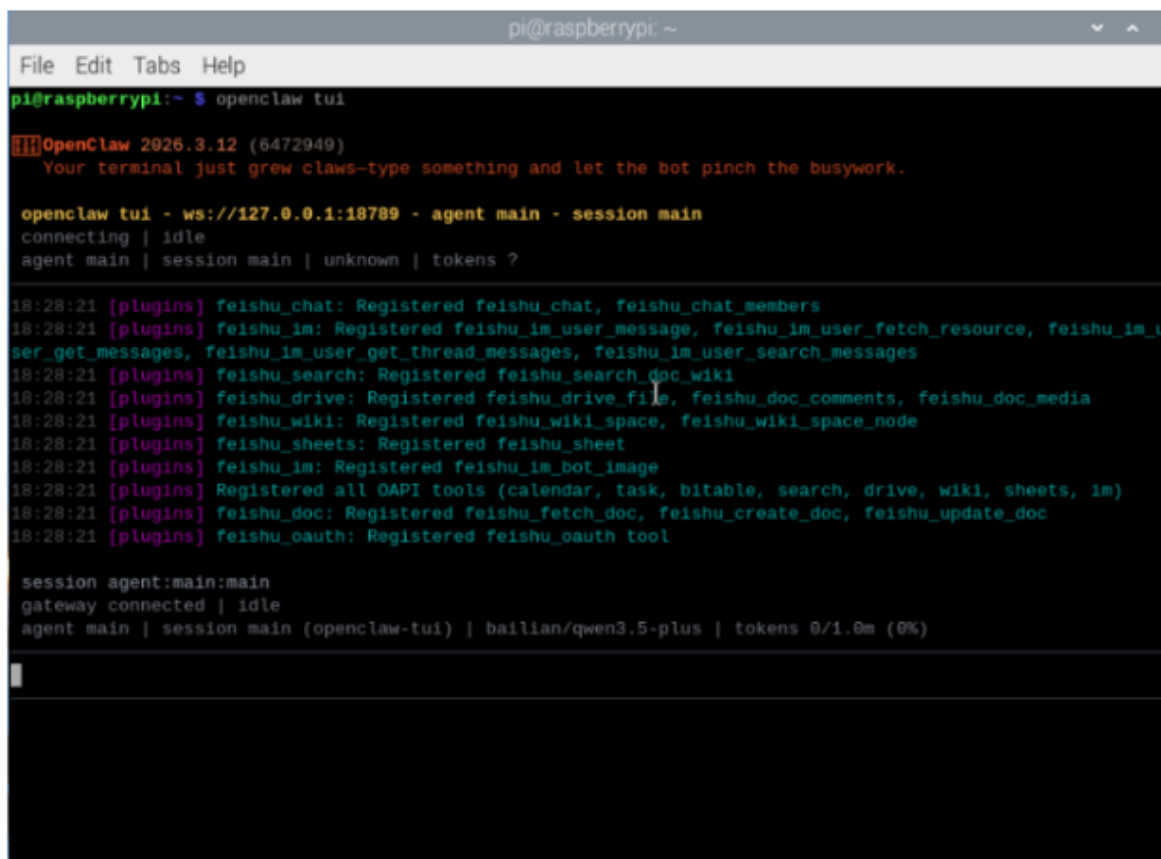
- Method 1: Generate script and run immediately
- Method 2: Only generate script, save to desktop first, then modify or run again later

4. Testing Script Generation Capability

Before starting this tutorial, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After confirming OpenClaw can respond normally, proceed with the script generation-specific test.

Enter TUI by typing the following in the terminal:

```
openclaw tui
```



```

pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $ openclaw tui
[11] OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

```

It is recommended to start with the simplest script requirements, for example:

- Help me write a script that makes RGB red lights flash 5 times, save to desktop and run
- Generate a script that reads temperature and humidity, don't run it yet
- Write a script that makes servo 1 swing back and forth between 0, 90, 180 degrees
- Generate a script that displays "Hello OpenClaw" on OLED

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote requirement sending, classroom demonstration	Can directly describe "help me write what script"	Used to demonstrate natural language generation capability
TUI	Local debugging, verify generation results	Easiest to see script generation and run returns	First choice for debugging solution
Voice Code	Voice interaction, complete AI experience	Can directly verbally describe script requirements	As the complete demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

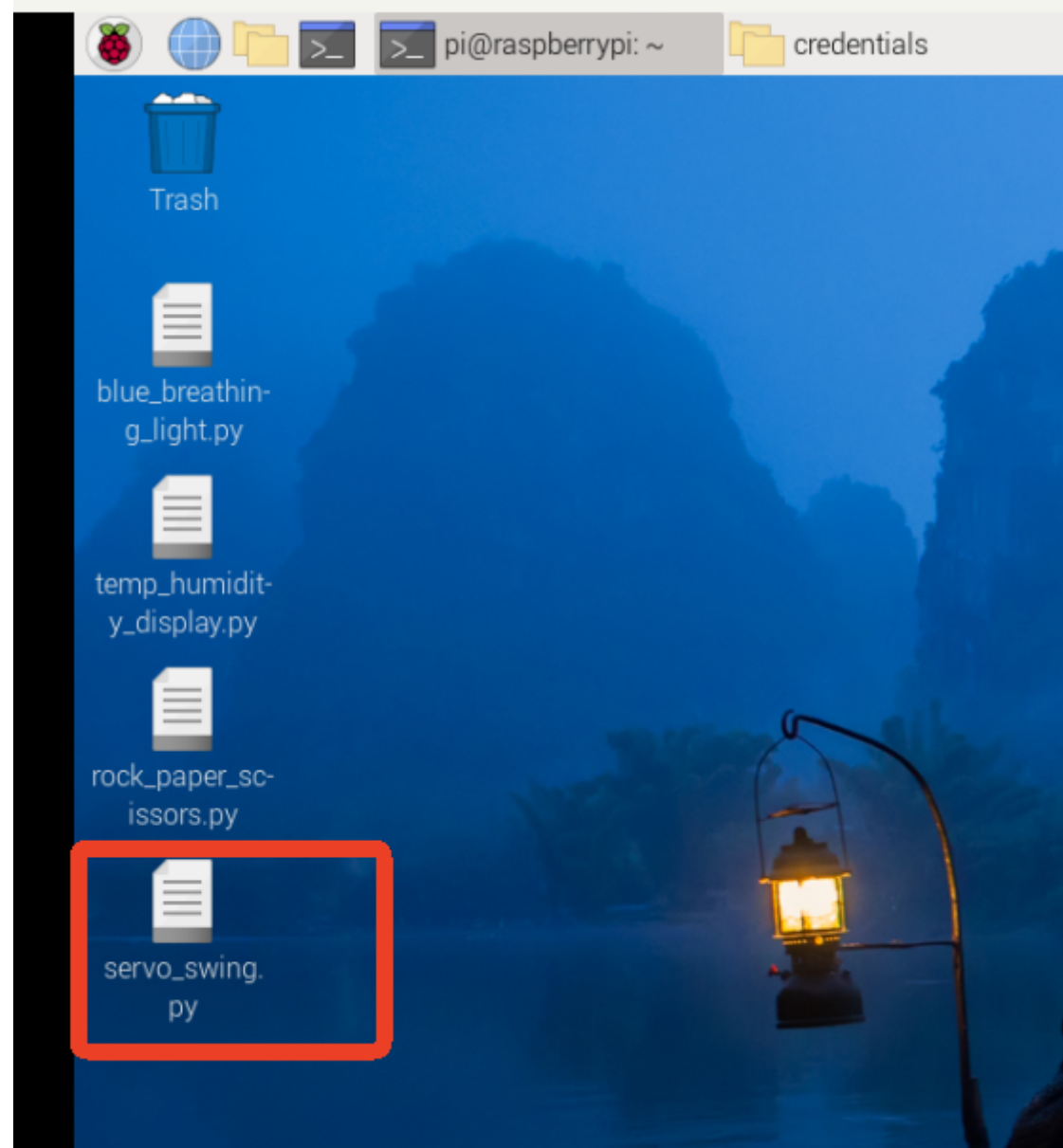
```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration steps, please refer directly to [05-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the script generation tutorial.

You can converse directly, for example:

- Generate a script that makes servo swing left and right for 10 seconds and run it



20:07



< 2



+86 153 3885 7526 (你)

给自己发消息

HTTP 400: Access denied, please make sure your account is in good standing. For details, see: <https://help.aliyun.com/zh/model-studio/error-code#overdue-payment>

19:59 ✓✓

Generate a script that makes servo swing left and right for 10 seconds and run it

20:07 ✓✓

Done! Generated `servo_swing.py` on the desktop and executed it successfully.

The servo swung left (0°) and right (180°) repeatedly for 10 seconds, then returned to the center position (90°).

20:07 ✓✓



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return



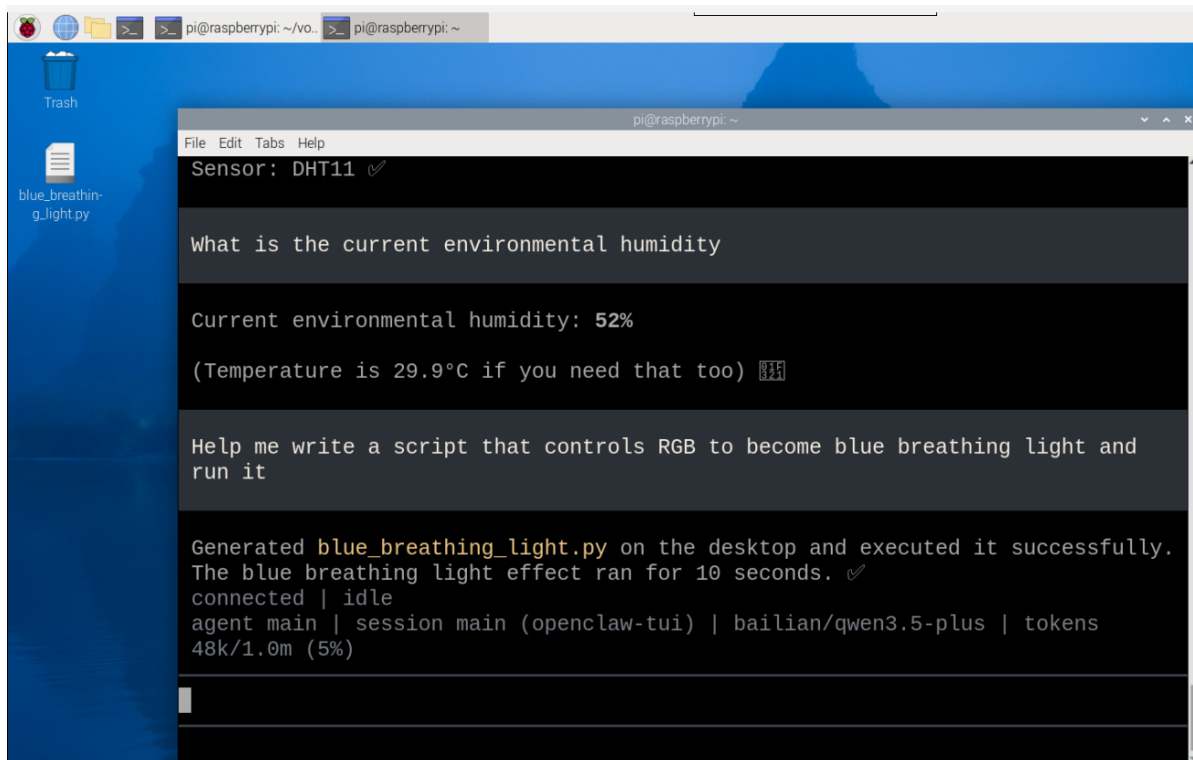
5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, simply type the following command in the terminal to continue the conversation:

```
openclaw tui
```

Then you can directly input script requirements, for example:

- Help me write a script that controls RGB to become blue breathing light and run it

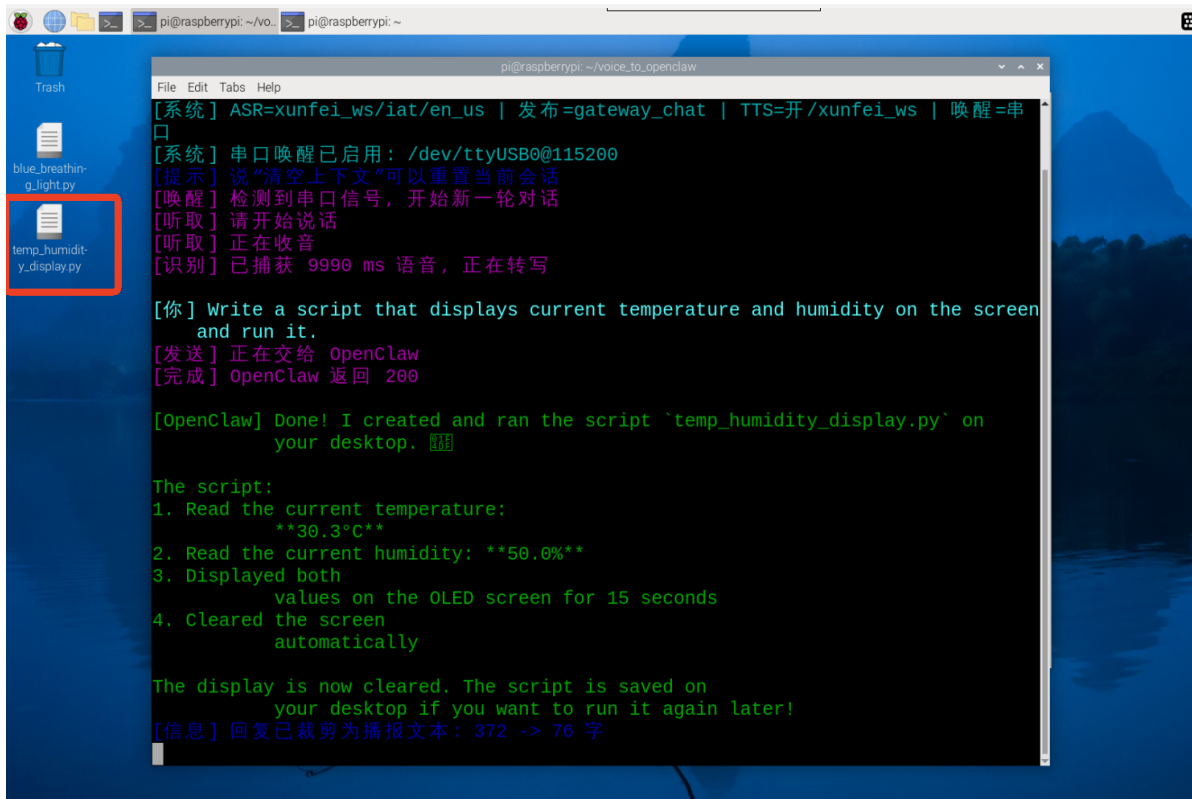


5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- Write a script that displays current temperature and humidity on the screen and run it



6. Comprehensive Cases

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Case 1: RGB Flashing Script

Experiment Objective

Have OpenClaw generate a basic lighting script to verify the "script creation + peripheral execution" chain.

Effect Description

OpenClaw will organize your natural language requirements into a Python script. After the script is saved to the desktop, it can be run directly or kept for later modification.

Example Commands

- Help me write a script that makes RGB red lights flash 5 times
- Generate a blue breathing light script, save to desktop and run
- Write a script that makes RGB do a rainbow light effect

6.2 Case 2: Servo Action Script

Experiment Objective

Generate a servo action script to solidify single conversation actions into repeatable programs.

Effect Description

This type of script is suitable for actions like "nodding", "head shaking", "swinging back and forth", "returning to center", and is also suitable for later expansion into classroom demonstrations or combined projects.

Example Commands

- Write a script that makes servo 1 move back and forth between 0, 90, 180 degrees
- Generate a servo nodding script
- Help me write a script that makes four servos return to 90 degrees together

6.3 Case 3: Temperature and Humidity Reading Script

Experiment Objective

Generate a script that reads current temperature and humidity and outputs results.

Effect Description

This type of script is suitable for sensor self-check, data sampling, or subsequent log recording, and is also the most commonly used type of basic script when doing scheduled tasks later.

Example Commands

- Generate a script that reads temperature and humidity, save to desktop
- Help me write a script that prints current temperature and humidity
- Write a script that reads DHT data once and runs it

6.4 Case 4: OLED Display Script

Experiment Objective

Generate a script that displays text or metric content on OLED.

Effect Description

This case is suitable for welcome pages, status pages, simple dashboards, and is also the foundation for later linking with capabilities like temperature and humidity and cameras.

Example Commands

- Help me write a script that displays "Hello OpenClaw" on OLED
- Generate a script that displays temperature and humidity on OLED
- Write a script that displays a page of simple status information

6.5 Case 5: Combined Script

Experiment Objective

Have OpenClaw generate a comprehensive script containing multiple peripheral actions at once.

Effect Description

This type of script can chain "read sensor", "display on OLED", "control RGB", "drive servo", suitable for transitioning from single peripheral control to small project scripts.

Example Commands

- Write a comprehensive script: read temperature and humidity and display on OLED
- Generate a script: first light up green light, then make servo rotate to 90 degrees
- Help me write a script: if temperature is too high, light up red light and prompt on OLED